Introduction to Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed.

There are three main types of machine learning:
1. Supervised Learning - Uses labeled data. Examples: Email spam detection, house price prediction.
2. Unsupervised Learning - Uses unlabeled data. Examples: Customer segmentation, clustering.
3. Reinforcement Learning - Learns through rewards and penalties. Examples: Game-playing AI, robotics.

Machine Learning is widely used in healthcare, finance, cybersecurity, and recommendation systems.

Deep Learning and Neural Networks

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with multiple layers.

Key Concepts:
- Input Layer
- Hidden Layers
- Output Layer

Popular Deep Learning frameworks:
- TensorFlow
- PyTorch
- Keras

Applications include Image Recognition, Speech Recognition, and Natural Language Processing.

Generative AI and Large Language Models

Generative AI focuses on creating new content such as text, images, audio, and code.

Large Language Models (LLMs) are trained on massive datasets and can generate human-like text.

Examples: GPT, Claude, Gemini, Llama.

Important Concepts:
• Prompt Engineering
• Retrieval Augmented Generation (RAG)
• Fine-Tuning
• AI Agents

RAG combines information retrieval with language generation to provide accurate and context-aware responses.